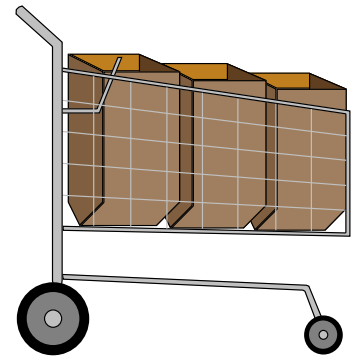


Association Rule Mining

Association Rule Mining

- Consider shopping cart filled with several items
- Market basket analysis tries to answer the following questions:
 - Who makes purchases?
 - What do customers buy together?
 - In what order do customers purchase items?

Market Basket Analysis



- INPUT: list of purchases by purchaser
 - do not have names
- *identify purchase patterns*
 - what items tend to be purchased together
 - obvious: steak-potatoes; beer-pretzels
 - what items are purchased sequentially
 - obvious: house-furniture; car-tires
 - what items tend to be purchased by season

Association Rules

- Categorize customer purchase behavior
- identify *actionable* information
 - purchase profiles
 - profitability of each purchase profile
 - use for marketing
 - layout or catalogs
 - select products for promotion
 - space allocation, product placement

Association Rules

- ***Benefits***
 - selection of **promotions, merchandising strategy**
 - sensitive to price: Italian entrees, pizza, pies, Oriental entrees, orange juice
 - uncover **consumer spending patterns**
 - correlations: orange juice & waffles
 - **joint promotional opportunities**

Applications

- Retail outlets
- Telecommunications
- Banks
- Insurance
 - link analysis for fraud
- Medical
 - symptom analysis

Purchase Profiles

- Each profile has an average profit per basket
 - Kids' fashion \$15.24 **push these**
 - Men's fashion \$13.41
 -
 - Smoker \$2.88 *don't push*
 - Student/home office \$2.55 *these*

Market Basket Analysis

- ***Affinity Positioning***
 - coffee, coffee makers in close proximity
- ***Cross-Selling***
 - cold medicines, kleenex, orange juice

Association Rules

- Wal-Mart customers who purchase Barbie dolls have a 60% likelihood of also purchasing one of three types of candy bars [*Forbes*, Sept 8, 1997]
- Customers who purchase maintenance agreements are very likely to purchase large appliances (Linoff and Berry experience)
- When a new hardware store opens, one of the most commonly sold items is toilet bowl cleaners (Linoff and Berry experience)

How can Association Rules be used?

- Let the rule discovered be

$\{\text{Bagels}, \dots\} \rightarrow \{\text{Potato Chips}\}$



- **Potato chips as consequent** => Can be used to determine what should be done to boost its sales
- **Bagels in the antecedent** => Can be used to see which products would be affected if the store discontinues selling bagels
- **Bagels in antecedent and Potato chips in the consequent** => Can be used to see what products should be sold with Bagels to promote sale of Potato Chips

What Is Association Rule Mining

■ Rule form

Antecedent $\rightarrow$ Consequent

■ Examples

- $\text{buys}(x, \text{"computer"}) \rightarrow \text{buys}(x, \text{"financial management software"})$
- $\text{age}(x, \text{"30..39"}) \wedge \text{income}(x, \text{"42..48K"}) \rightarrow \text{buys}(x, \text{"car"})$

Definition: Frequent Itemset

- **Itemset**

- A collection of one or more items
 - Example: {Milk, Bread, Diaper}
- k-itemset
 - An itemset that contains k items

- **Support count (σ)**

- Frequency of occurrence of an itemset
- E.g. $\sigma(\{\text{Milk, Bread, Diaper}\}) = 2$

- **Support**

- Fraction of transactions that contain an itemset
- E.g. $s(\{\text{Milk, Bread, Diaper}\}) = 2/5$

- **Frequent Itemset**

- An itemset whose support is greater than or equal to a *minsup* threshold

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Definition: Association Rule

- **Association Rule**

- An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
- Example:
 $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- **Rule Evaluation Metrics**

- Support (s)
 - ◆ Fraction of transactions that contain both X and Y
- Confidence (c)
 - ◆ Measures how often items in Y appear in transactions that contain X

Example:

$\{\text{Milk, Diaper}\} \Rightarrow \text{Beer}$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

Association Rule

Association rules are rules presenting association or correlation between itemsets.

$$\text{support}(A \Rightarrow B) = P(A \cup B)$$

$$\begin{aligned}\text{confidence}(A \Rightarrow B) &= P(B/A) \\ &= \frac{P(A \cup B)}{P(A)}\end{aligned}$$

$$\begin{aligned}\text{lift}(A \Rightarrow B) &= \frac{\text{confidence}(A \Rightarrow B)}{P(B)} \\ &= \frac{P(A \cup B)}{P(A)P(B)}\end{aligned}$$

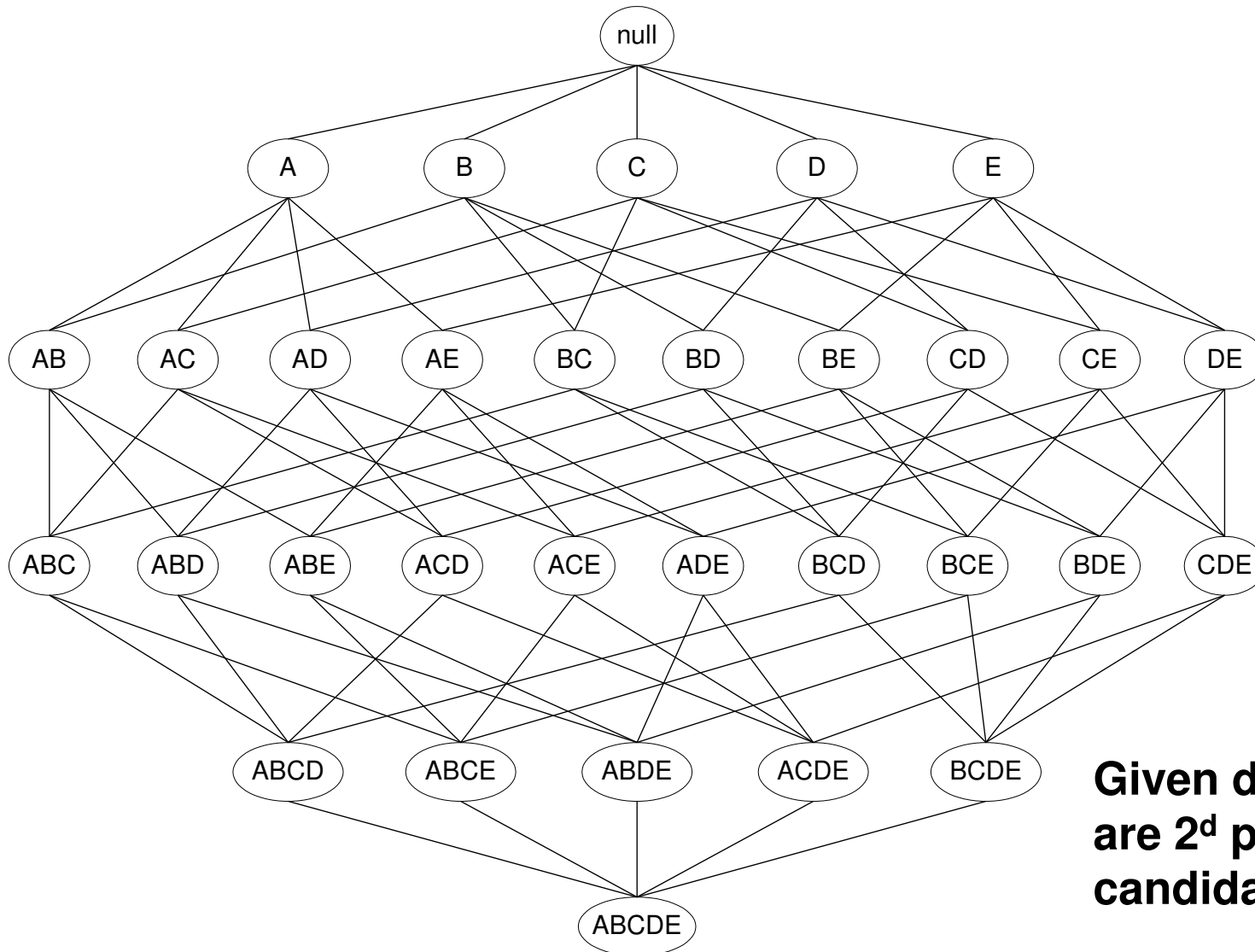
where $P(A)$ is the percentage (or probability) of cases containing A .

Frequent Itemsets Mining

TID	Transactions
100	{ A, B, E }
200	{ B, D }
300	{ A, B, E }
400	{ A, C }
500	{ B, C }
600	{ A, C }
700	{ A, B }
800	{ A, B, C, E }
900	{ A, B, C }
1000	{ A, C, E }

- Minimum support level 50%
 - {A},{B},{C},{A,B}, {A,C}

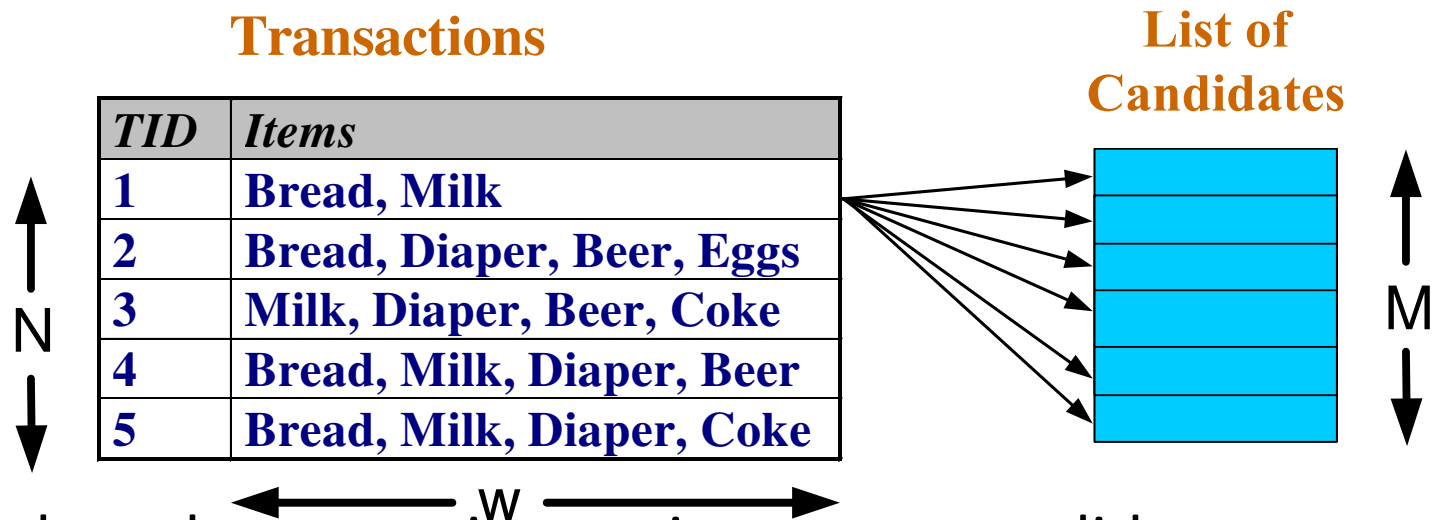
Frequent Itemset Generation



Given d items, there are 2^d possible candidate itemsets

Frequent Itemset Generation

- Brute-force approach:
 - Each itemset in the lattice is a **candidate** frequent itemset
 - Count the support of each candidate by scanning the database

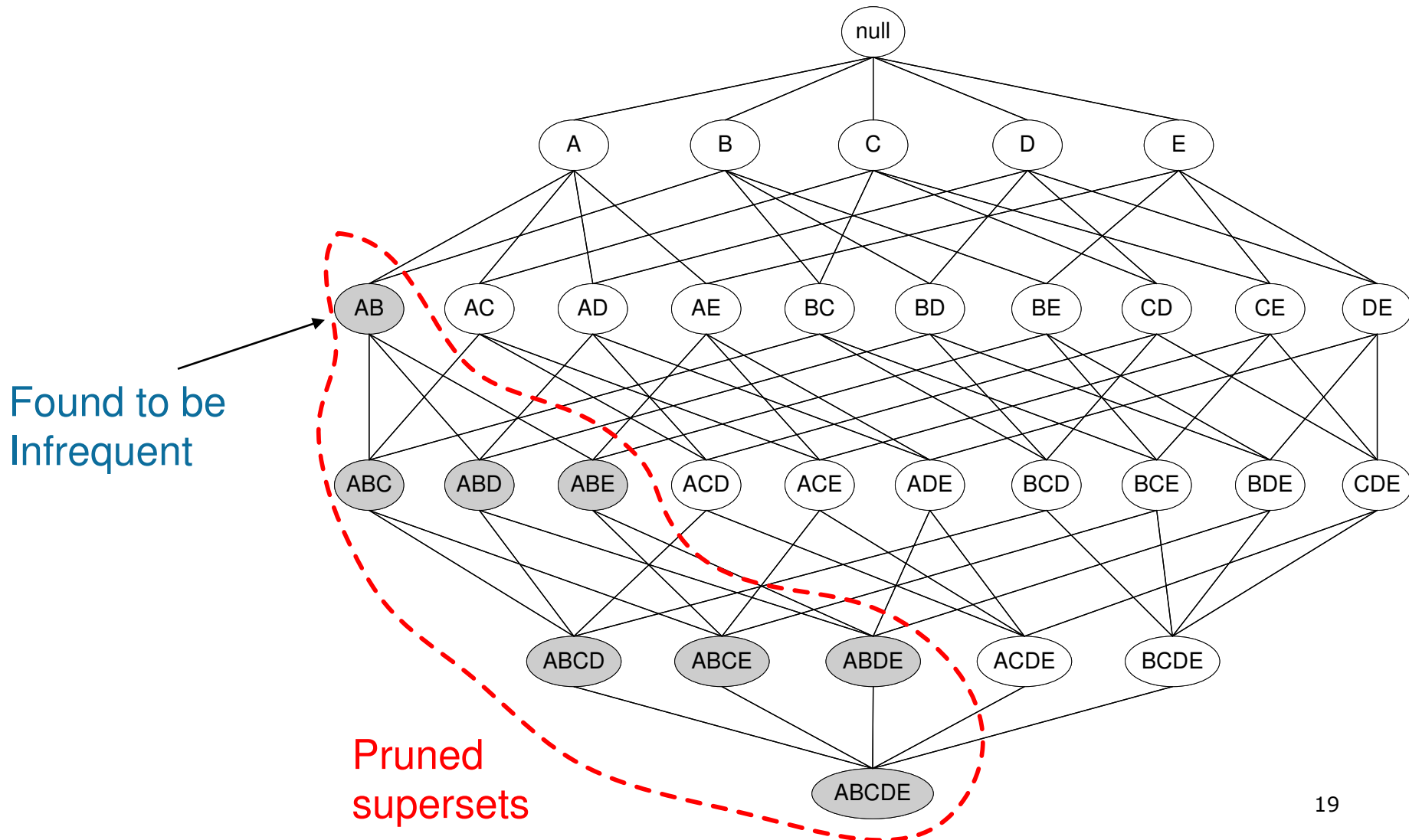


- Match each transaction against every candidate
- Complexity $\sim O(NMw) \Rightarrow$ **Expensive since $M = 2^d$!!!**

Reducing Number of Candidates

- **Apriori principle:**
 - If an itemset is frequent, then all of its subsets must also be frequent
- Apriori principle holds due to the following property of the support measure:
$$\forall X, Y : (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$
 - Support of an itemset never exceeds the support of its subsets
 - This is known as the **anti-monotone** property of support

Illustrating Apriori Principle



Illustrating Apriori Principle

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,
 ${}^6C_1 + {}^6C_2 + {}^6C_3 = 41$
 With support-based pruning,
 $6 + 6 + 1 = 13$



Itemset	Count
{Bread,Milk,Diaper}	3

Triplets (3-itemsets)



Challenges of Frequent Itemset Mining

- Challenges
 - Multiple scans of transaction database
 - Huge number of candidates
 - Tedious workload of support counting for candidates
- Improving Apriori: general ideas
 - Reduce passes of transaction database scans
 - Shrink number of candidates
 - Facilitate support counting of candidates

Alternative Methods for Frequent Itemset Generation

- Representation of Database
 - horizontal vs vertical data layout

Horizontal
Data Layout

TID	Items
1	A,B,E
2	B,C,D
3	C,E
4	A,C,D
5	A,B,C,D
6	A,E
7	A,B
8	A,B,C
9	A,C,D
10	B

Vertical Data Layout

A	B	C	D	E
1	1	2	2	1
4	2	3	4	3
5	5	4	5	6
6	7	8	9	
7	8	9		
8	10			
9				

ECLAT

- For each item, store a list of transaction ids (tids)

Horizontal
Data Layout

TID	Items
1	A,B,E
2	B,C,D
3	C,E
4	A,C,D
5	A,B,C,D
6	A,E
7	A,B
8	A,B,C
9	A,C,D
10	B

Vertical Data Layout

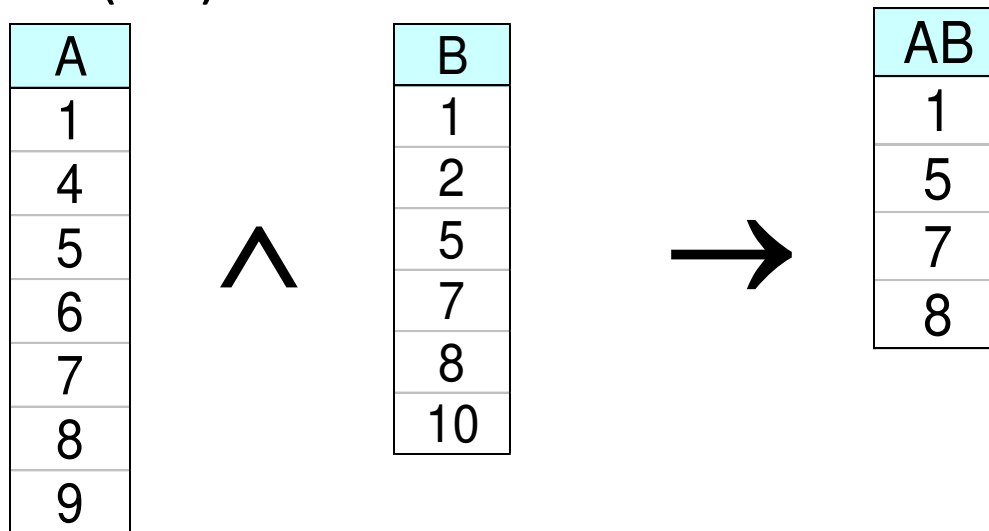
A	B	C	D	E
1	1	2	2	1
4	2	3	4	3
5	5	4	5	6
6	7	8	9	
7	8	9		
8	10			
9				



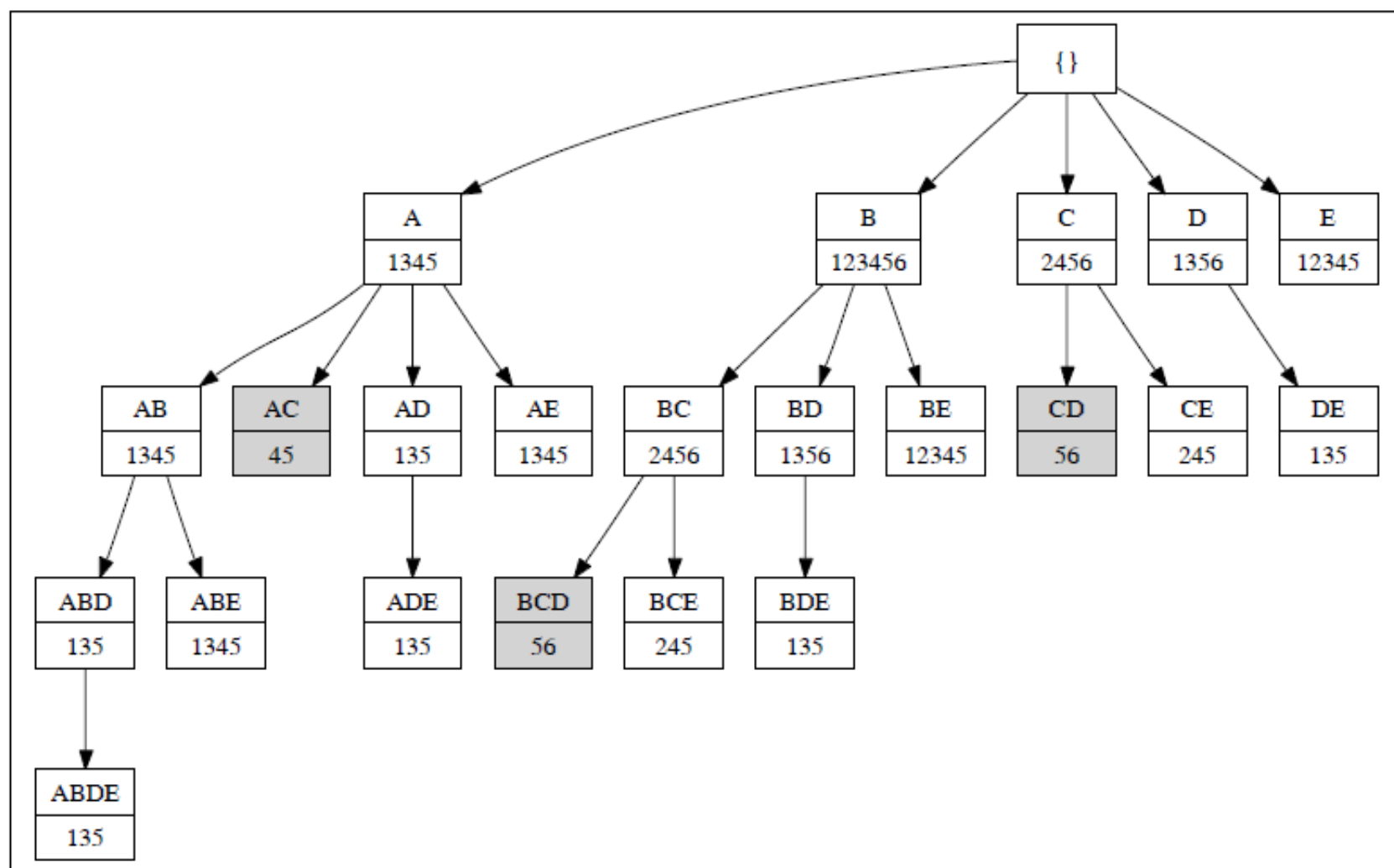
TID-list

ECLAT

- Determine support of any k-itemset by intersecting tid-lists of two of its (k-1) subsets.

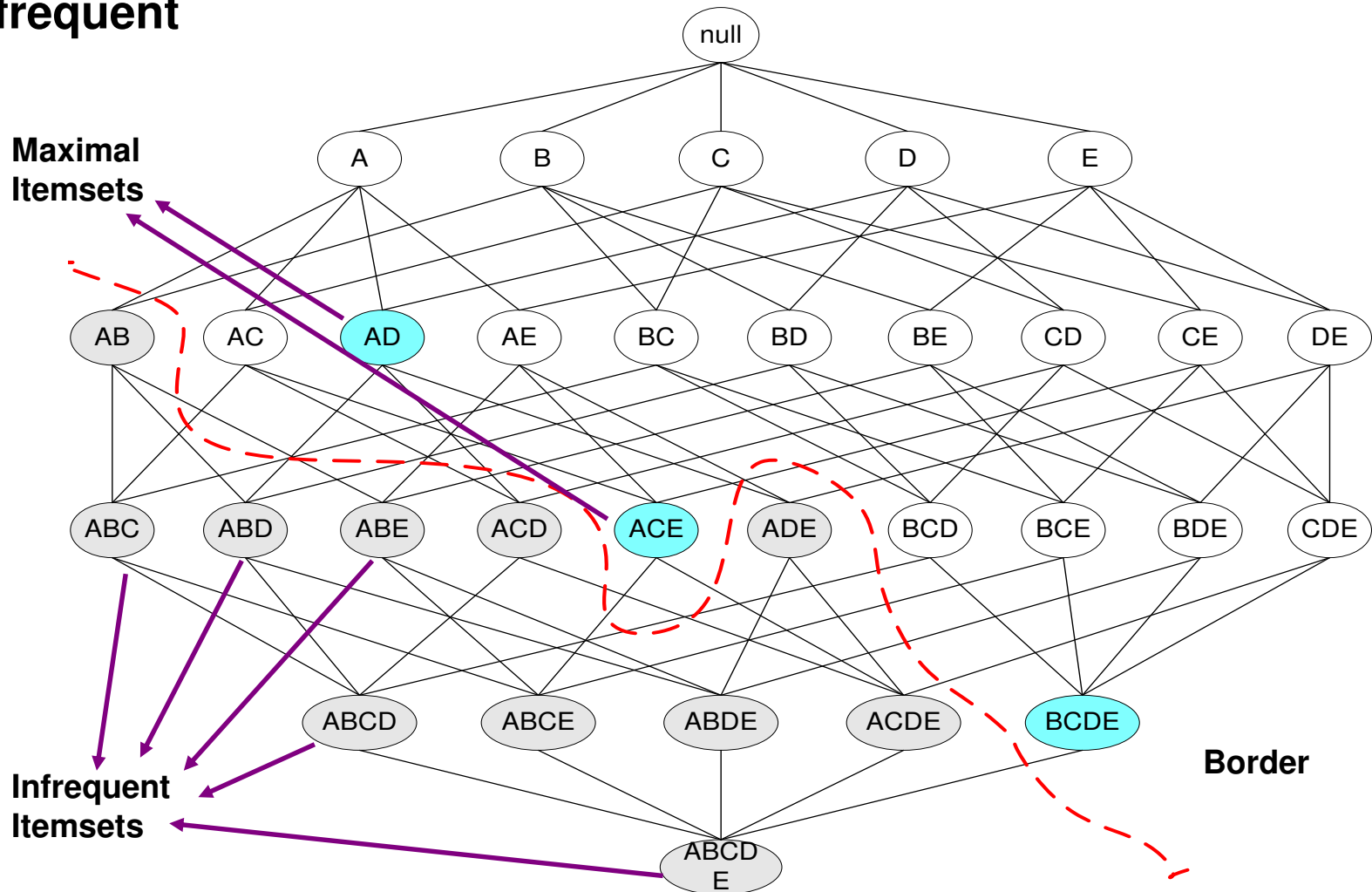


- 3 traversal approaches:
 - top-down, bottom-up and hybrid
- Advantage: very fast support counting
- Disadvantage: intermediate tid-lists may become too large for memory



Maximal Frequent Itemset

An itemset is maximal frequent if none of its immediate supersets is frequent



Closed Itemset

- An itemset is closed if none of its immediate supersets has the same support as the itemset

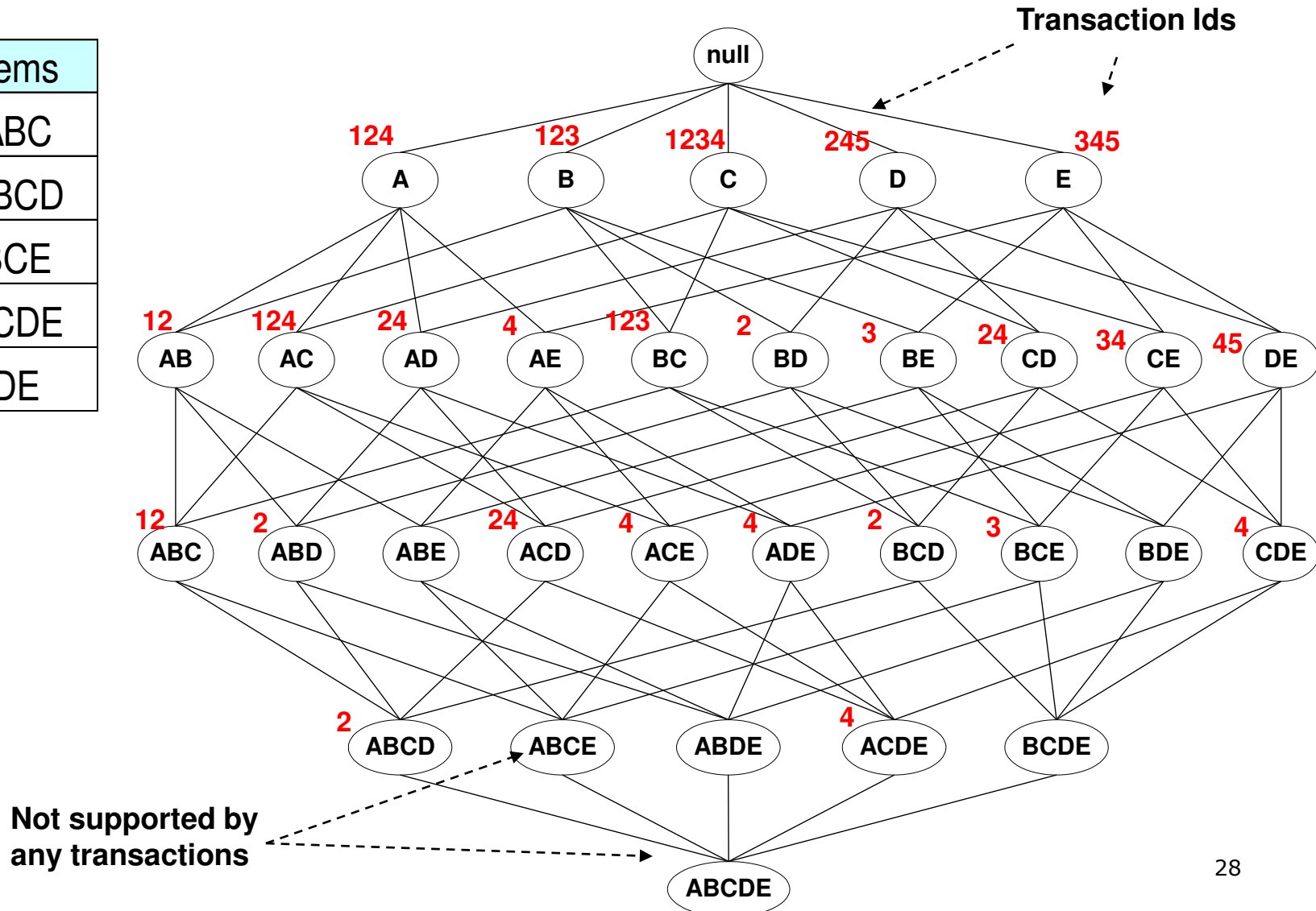
TID	Items
1	{A,B}
2	{B,C,D}
3	{A,B,C,D}
4	{A,B,D}
5	{A,B,C,D}

Itemset	Support
{A}	4
{B}	5
{C}	3
{D}	4
{A,B}	4
{A,C}	2
{A,D}	3
{B,C}	3
{B,D}	4
{C,D}	3

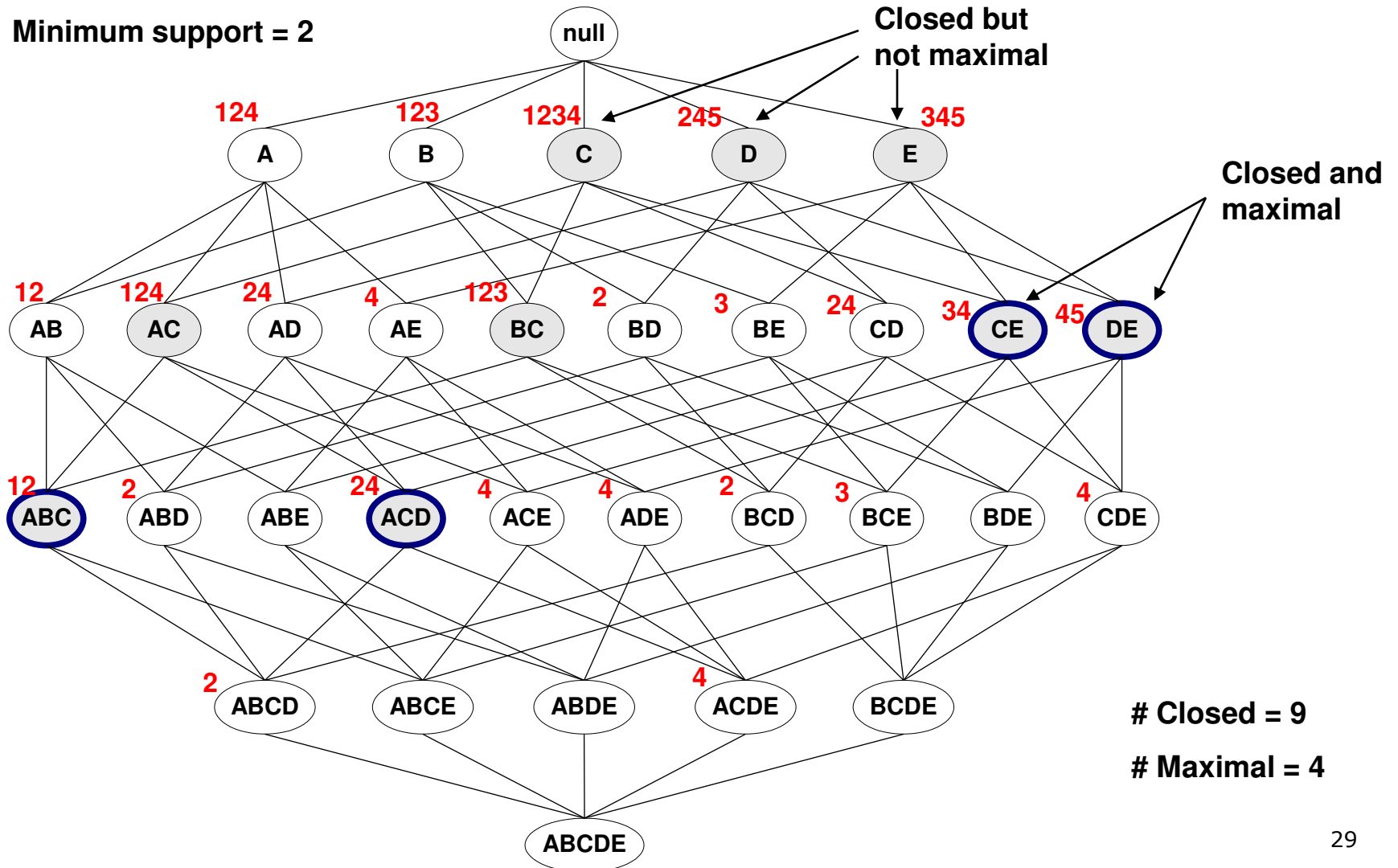
Itemset	Support
{A,B,C}	2
{A,B,D}	3
{A,C,D}	2
{B,C,D}	3
{A,B,C,D}	2

Maximal vs Closed Itemsets

TID	Items
1	ABC
2	ABCD
3	BCE
4	ACDE
5	DE



Maximal vs Closed Frequent Itemsets



Maximal vs Closed Itemsets

